Control Engineering Practice 72 (2018) 138–150

Contents lists available at ScienceDirect

Control Engineering Practice

journal homepage: www.elsevier.com/locate/conengprac

Energy saving control in separate meter in and separate meter out control system ☆

Guangrong Chen, Junzheng Wang, Shoukun Wang, Jiangbo Zhao, Wei Shen *

Key Laboratory of Intelligent Control and Decision of Complex Systems, Beijing Institute of Technology, Beijing, 100081, China

A R T I C L E I N F O

Keywords:
SMISMO
Indirect adaptive robust control
Dynamic surface control
Fast parameter estimation
Energy saving
Grey predictor
Disturbance observer

A B S T R A C T

With the demand for energy efficiency in electro-hydraulic servo system (EHSS) increasing, the separate meter in and separate meter out (SMISMO) control system draws massive attention. In this paper, the SMISMO control system was decoupled completely into two subsystems by the proposed indirect adaptive robust dynamic surface control (IARDSC) method. Besides, a fast parameter estimation scheme was proposed to adapt to the parameter change for a better estimation performance. Also, a supply pressure controller with a disturbance observer and a supply flow rate controller with a grey model predictor were investigated and employed to save the power consumption. Finally, experimental results showed that the proposed IARDSC could achieve a good trajectory tracking performance with the fast parameter estimation. Meanwhile, the two energy saving techniques were validated.

1. Introduction

As applications of EHSS become more and more popular, the demand for low cost, high-level control performance and significant energy saving schemes get stronger and stronger. Generally, the control performance can be seen as the fundamental index in this kind of systems and many control algorithms have been issued (Chen et al. 2016b; Guan & Pan, 2008; Yao, Bu, & Chiu, 2001; Yao, Bu, Reedy, & Chiu, 2000). As for energy saving, the hydraulic energy E from t_0 to t_1 can be defined as:

$$E = \int_{t_0}^{t_1} P_s(\tau)Q_s(\tau)d\tau \tag{1}$$

where P_s is the fluid source supply pressure and Q_s is the fluid source supply flow rate. Obviously, two ways can be utilized to reduce the usage of energy:

- Reducing the fluid source supply pressure $P_s(t)$.
- Reducing the fluid source supply flow rate $Q_s(t)$.

On one hand, only taking reducing the fluid source supply pressure into consideration, pressures at the two cylinder chambers are desired to be as low as possible when a certain pressure difference is kept to maintain the motion task. Thus, independent control of two chamber pressures is one way to save energy. On the other hand, only considering reducing the fluid source supply flow rate, the fluid source is required to provide enough flow rate to maintain the given motion trajectory of load. Thus, reducing the supply flow rate appropriately is another way to decrease the power consumption.

For this issue of energy saving, many configurations have been employed in EHSS, such as mobile hydraulic valve, load sensing (Breeden, 1981) and the proposed SMISMO control systems (Jansson & Palmberg, 1990). Eliminating the mechanical linkage between the meter-in and meter-out orifices is a well known technique and has been used in hydraulic industry for several years. For example, Liu and Yao (2006, 2008), Yao and DeBoer (2002) and Yao and Liu, (2002) have done trajectory tracking control utilizing five high speed switch valves, and both good trajectory following precision and energy saving characteristic have been achieved. Liu, Xu, Yang, and Zeng (2009a), Liu, Xu, Yang, and Zeng (2009b) and Liu, Xu, Yang, and Zeng (2009c) have done some comparative simulations between SMISMO valve arrangement control systems and traditional proportional direction control systems, which showed the better energy saving characteristic in SMISMO control systems. Aardema (1996) used two directional control valves and (Chen, Wang, Wang, & Ma, 2016a) used two servo valves to do the same research: One valve controlled the chamber flows of head end and the other controlled the chamber flows of rod end. In addition, the usage of

☆ This work is supported by National Natural Science Foundation of China under Grant No. 51675041.
* Corresponding author.
E-mail address: sw_she@bit.edu.cn (W. Shen).

http://dx.doi.org/10.1016/j.conengprac.2017.09.001
Received 10 January 2017; Received in revised form 31 August 2017; Accepted 1 September 2017
Available online 20 December 2017
0967-0661/

four independent valves of either poppet type or one-way unidirectional type is a more common scheme, which makes sure that the meter-in and meter-out flow can be truly independently controlled. This scheme is used in many studies among the mobile hydraulics industry (Aardema & Koehler, 1999; Book & Goering, 1999; Chen et al., 2016a). In order to obtain the hardware capability of independent control of each meter-in and meter-out ports, the 'Independent Metering Valve' (Aardema & Koehler, 1999) or 'Smart Valve' (Book & Goering, 1999) is involved, which makes it available to control both cylinder states completely. When the hardware flexibility is properly utilized, the dual objectives of precise motion trajectory tracking control and high hydraulic energy efficiency can be achieved to some extent.

With the increasing demand for a better control performance, it is necessary to explicitly consider the effect of nonlinearities and uncertainties associated with the electro-hydraulic systems. As such, the design methodology 'integrator backstepping (IB)' has received a great deal of interest. The book by Krstic, Kanellakopoulos, and Kokotovic (1995) developed the backstepping approach to the point of a step by step design procedure. In the recent 20 years, adaptive robust control based on the idea of IB gains vast attention in many literatures, such as adaptive robust control (ARC) (Escareno, Rakotondrabe, & Habineza, 2015; Pazelli, Terra, & Siqueira, 2011; Yao & Tomizuka, 1997) and indirect adaptive robust control (IARC) (Yao & Palmer, 2002). Although an accurate parameter estimation is achieved in ARC&IARC, the estimation method is only applied to constant parameter situations. When the parameters are changing with respect to time, the poor estimation speed makes it difficult to obtain the true values of parameters and then results in poor trajectory tracking performance. Hence, a fast parameter estimation is needed and makes sense here.

Due to the basis of IB technique, the above methods suffer from the problem of 'explosion of terms', that is, the complexity of controller grows drastically as the order of the system increases. Swaroop, Hedrick, Yip, and Gerdes (2000) proposed a dynamic surface control technique to solve this problem by introducing a first-order filter into the synthesized virtual control law at each step of the backstepping design procedure. Literatures show that DSC technique is suitable to solve the 'explosion of terms' problem (Li, Chen, Gan, Fang, & Zhang, 2010; Na, Ren, Herrmann, & Qiao, 2011; Qiu, Liang, & Dai, 2015; Song, Zhang, Zhang, & Lu, 2014). Also, in Song et al. (2014) and Na et al. (2011), DSC was combined with robust and adaptive control to achieve guaranteed performance, respectively. However, when the system suffers from both parameter uncertainty and disturbance, DSC with either adaptation or robustness will fail to achieve a better performance. Therefore, in this paper, both adaptive control and robust control will be combined with DSC, so that the parametric uncertainty and unknown disturbance can be restrained at the same time. By utilizing DSC technique in the IARC design procedure, and with a construction of fast parameter estimation, an IARDSC with fast parameter estimation is proposed to achieve fast and accurate parameter estimation while maintaining a guaranteed performance and eliminating the 'explosion of terms' under parametric uncertainty and unknown disturbance.

The disturbance observer is first proposed in Chen (2003). For systems satisfying the matched conditions, the disturbance observer is a method that has been applied in conjunction with controller (Lu, 2009; Mohammadi, Tavakoli, Marquez, & Hashemzadeh, 2013; Sun, Li, & Lee, 2015). For mismatched nonlinear systems, many significant results with the disturbance observer approach has been investigated in the literature (Yang, Chen, & Li, 2011) recently. For example, Ginoya, Shendge, and Phadke (2014) and Yang, Li, and Yu (2013) proposed a novel sliding mode controller by using disturbance observer to counteract the influence of mismatched uncertainties with a new sliding surface including the estimate of unmatched disturbances. Moreover, it could alleviate the chatter problem in control substantially except for counteracting the influence of mismatched uncertainties. Also, the disturbance observer is utilized in generalized extended state observer based control (Yao, Jiao, & Ma, 2014a, 2014b). Thus, in the SMISMO control system, the disturbance observer can be utilized to estimate the load force, which is taken as a reference to control the fluid source supply pressure for saving energy.

Due to the existence of the pipeline between the pump and valve, there is a delay and drop in the supply flow rate and pressure. Thus, a predictor is needed and a grey model predictor is a suitable choice here. The concept of grey systems is originally developed by Deng (Julong, 1989), and the grey theory is famous for its ability to tackle systems with partially unknown parameters. The technique of grey prediction has been successfully employed to deal with many engineering problems, such as hydraulic system control (Chen, Wang, Ma, & Hao, 2015; Chiang & Tseng, 2004). In grey system theory, a grey prediction model is one of the most important parts, and one core of grey prediction models is GM(1,1) model (Zeng, Liu, & Xie, 2010). Therefore, the GM(1,1) model is employed to predict the motion of load, which can be utilized to control the fluid source supply flow rate for reducing the power consumption.

Based on our published work (Chen et al., 2015), this paper focus on the further research about energy saving control in EHSSs without losing of tracking performance. The main contributions are concluded as follows:

- Our previous proposed study, IARDSC (Chen, Wang, Wang, Zhao, & Shen, 2017) and fast parameter estimation (Hao, Wang, Zhao, & Wang, 2016) were employed to maintain the tracking precision since the SMISMO control system is with internal parameter uncertainties, external disturbances, and the influence of the following energy saving control.
- Two energy saving techniques: reducing the supply pressure by using a disturbance observer and reducing the supply flow rate by using a grey model predictor, were proposed and analyzed.
- To validate the effectiveness of proposed method, comparative experiments were implemented and experimental results showed a significant energy saving performance with the required tracking performance guaranteed.

This paper is organized as follows. The SMISMO control system is modeled in Section 2. In Section 3, the IARDSC and a fast parameter estimation algorithm are proposed for the SMISMO control system. Section 4 gives out the details about two main ways to save energy: reducing the fluid source supply pressure via a load observer to estimate the proper pump pressure and reducing the fluid source supply flow rate by using a grey model to predict the flow rate demand of load. Experimental results are presented in Section 5 to show the effectiveness of the proposed method. Conclusions are drawn in Section 6. Moreover, the related proof is analyzed in the Appendix.

2. System modeling

The SMISMO control system scheme considered here is shown in Fig. 1. This system is mainly composed of a hydraulic cylinder with an inertia load, two proportional directional control valves (PDCV1 & PDCV2), an electro-hydraulic proportional relief valve and the fluid source. The relief valve is intended to control the supply pressure, which is proportional to its control voltage input, while the fluid source, driven by a servo motor, is utilized to control the supply flow rate. Noting that the two PDCVs have different rated flow rates because of the asymmetry of single-rod cylinder. The rated flow rate of the PDCV on the cylinder-end side is larger than that on the rod-end side. Actually, the whole system is a multiple input multiple output (MIMO) system, which can be divided into four single input single output (SISO) subsystems based on their own control input: a motion servo system (control voltage of one PDCV), a backpressure regulating system (control voltage of the other PDCV), a supply pressure control system (control voltage of relief valve), and a supply flow rate control system (control voltage of servo motor). However, every subsystem is not independent of others and couples together, especially the former two subsystems. Therefore, they

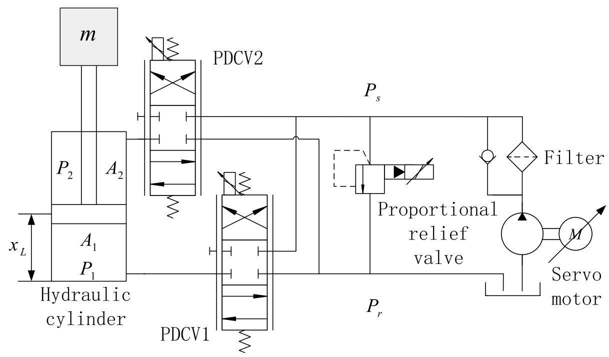

Fig. 1. The SMISMO control system.

cannot be investigated separately. There are two main objectives in this paper. One is to design two available control laws for the control inputs of two PDCVs such that the output position x_L can track the desired trajectory x_{Ld} as closely as possible and make way for reducing power consumption as well. The other objective is to design the relief valve controller and the servo motor controller to achieve energy saving by referring to (1).

Define $P_1A_1 - P_2A_2 = P_L$ as the load force to drive the cylinder rod. For tracking the given motion trajectory precisely, P_L should be accurately controlled to generate the required load force that can accomplish the desired motion. If only a single conventional four-way valve is employed, the two chamber pressures P_1 and P_2 cannot be controlled independently because they are coupled with each other. Thus, the solution is unique in this case. However, the solution is not unique any more if two PDCVs are utilized. That is because both P_1 and P_2 can be independently controlled, which results in a tremendous flexibility in controlling the system. The purpose here is regulating P_L to track the desired motion trajectory of load while keep P_1 and P_2 or the backpressure as low as possible for saving energy. Therefore, the pressure of one of the two cylinder chambers (backpressure) should be kept at a low value, which is treated as off-side, while the pressure of the other chamber would be responsible for the motion tracking and is treated as working-side in the following. Furthermore, in the hydraulic industry, the working mode selection is depending on given tasks and done normally based on the desired motion and the motion direction of the load only (Liu & Yao, 2002).

The dual objectives of this study are as follows:

- Energy usage (main). The primary objective is to minimize the overall energy usage without too much sacrifice of achievable performance. For example, regulating the off-side pressure to track the desired backpressure, especially when the value of backpressure is constant.
- Tracking performance (auxiliary). Given the desired motion trajectory x_{Ld}, the secondary objective is to synthesize control signals for the two PDCVs such that the output x_L tracks any specified reference x_{Ld} as closely as possible in spite of various model uncertainties and disturbances.

According to the Newton's Law, the load dynamics is described by

$$P_1A_1 - P_2A_2 = m\ddot{x}_L + F_f + F + \Delta f \tag{2}$$

where P_1 and P_2 are pressures inside the two chambers of the cylinder; A_1 and A_2 are effective areas of the two chambers; m and x_L are the mass and displacement of load respectively; F_f is the modeled friction force; F is the constant part of load force; Δf is the combined uncertainty due to variable system load, external disturbances, unmodeled friction forces and hard-to-model terms.

According to the flow rate continuity, the cylinder dynamics can be written as:

$$\begin{aligned} \frac{V_1}{\beta_e}\dot{P}_1 &= Q_1 - A_1\dot{x}_L + \Delta Q_{10} \\ \frac{V_2}{\beta_e}\dot{P}_2 &= Q_2 + A_2\dot{x}_L + \Delta Q_{20} \end{aligned} \tag{3}$$

where $V_1 = V_{10} + A_1x_L$, which represents the total control volume of the first chamber; $V_2 = V_{20} - A_2x_L$, which represents the total control volume of the second chamber; V_{10} and V_{20} represent the two chamber volumes when x_L=0; β_e is the effective bulk modulus; ΔQ_{10} and ΔQ_{20} are the disturbance fluid flow including the internal leakage and external leakage; Q_1 represents supply flow rate to the forward (or cylinder-end) chamber; Q_2 represents ingress flow rate of the return (or rod-end) chamber. Thus, Q_1 and Q_2 are the orifice flows through PDCV1 and PDCV2 respectively and can be described by

$$Q_i = f_i(x_{vi}, \Delta P_i), \Delta P_i = \begin{cases} P_s - P_i, x_{vi} > 0 \\ P_i - P_r, x_{vi} < 0 \end{cases}, i = 1,2 \tag{4}$$

where P_s represents supply pressure of the fluid, P_r represents tank or reference pressure, $f_i(x_{vi}, \Delta P_i)$ is the nonlinear orifice flow mapping function of the orifice opening x_{vi} and the pressure drop ΔP_i.

Regardless of the valve dynamics, the servo valve can be modeled as a static function mapping the input signal and the pressure drop across the valve into the metered flow rate. Thus, $f_i(x_{vi}, \Delta P_i)$ can be described as

$$f_i(x_{vi}, \Delta P_i) = C_d W x_{vi}\sqrt{\frac{2}{\rho}\Delta P_i} \tag{5}$$

where C_d is the discharge coefficient; W is the valve spool area gradient; ρ is the fluid mass density.

In fact, the nonlinear flow mapping $f_i(x_{vi}, \Delta P_i)$ can never be perfect. Then, Q_i should be expressed as:

$$Q_i = Q_{iM} + \tilde{Q}_i, i = 1,2 \tag{6}$$

where Q_{iM} is the flow obtained from the valve flow mappings, which implies $Q_{iM} = f_i(x_{vi}, \Delta P_i)$ and $\tilde{Q}_i$ is the modeling errors of the flow mappings, which would be eliminated by robust feedback.

Generally, the valve control voltage u_{vi} is related to x_{vi} and can be considered as a second order transfer function given by

$$\frac{x_{vi}(s)}{u_{vi}(s)} = \frac{k_{vi}\omega_{vi}^2}{s^2 + 2\xi_{vi}\omega_{vi}s + \omega_{vi}^2} \tag{7}$$

Noting that the two PDCVs are configured with two different rated flow rates. According to the experiments, simulations, and data sheet, the gain, damping ratio, and natural frequency of PDCV1 are about $k_{v1} = 0.0097, \xi_{v1} = 0.7, \omega_{v1} = 86.2$ rad/s, while the gain, damping ratio, and natural frequency of PDCV2 are about $k_{v2} = 0.0065, \xi_{v2} = 0.68, \omega_{v2} = 91.4$ rad/s.

Compared with the conventional electro-hydraulic servo system using a single valve, the SMISMO control system using two valves is decoupled by the two valves in machinery. However, this kind of decoupling is partial, and there still exist coupled relations between the load displacement and the backpressure via the effect of internal state, such as the pressure and flow rate. In order to decouple the SMISMO control system completely, IARC method is introduced to deal with the problem. The main idea is defining the coupled relations as estimated parameters. Then, the adaptive algorithm is utilized to divide the SMISMO control system into two SISO systems: the working-side (in-let) system and off-side (out-let) system. The robust feedback is utilized to reject the effect of disturbances and uncertainties, and DSC is proposed in the IARC controller to simplify the design procedure, eliminate the 'explosion of terms', and decrease the computational cost. To facilitate, superscript 'w' represents the working-side system and 'o' represents the off-side system.

2.1. The working-side system

The objective of the working-side indirect adaptive robust motion controller is to make the load track the given trajectory x_{Ld}. To illustrate the design procedure, P_1 is taken as an example of the working-side. The controller design for P_2 follows the same procedure and is omitted.

Choose state variables as $x^w = [x_1^w \quad x_2^w \quad x_3^w \quad x_4^w \quad x_5^w]^{\mathrm{T}} \triangleq [x_L \quad \dot{x}_L \quad P_1 \quad x_{v1} \quad \dot{x}_{v1}]^{\mathrm{T}}$, the working-side system dynamic state function can be described as

$$\begin{cases} \dot{x}_1^w = x_2^w \\ \dot{x}_2^w = \dfrac{1}{m}(x_3^w A_1 - P_2 A_2 - F_f - F - \Delta f) \\ \dot{x}_3^w = \dfrac{\beta_e}{V_1}\left(C_d W x_4^w \sqrt{\dfrac{2}{\rho}\Delta P_1} - A_1 x_2^w - \Delta Q_1\right) \\ \dot{x}_4^w = x_5^w \\ \dot{x}_5^w = k_{v1}\omega_{v1}^2 u_{v1} - \omega_{v1}^2 x_4^w - 2\xi_{v1}\omega_{v1} x_5^w \end{cases} \tag{8}$$

where $\Delta Q_1 = \Delta Q_{10} + \tilde{Q}_1$, $P_2 A_2$ can be seen as a coupled item of working-side system with off-side system. To decouple it, IARC method is utilized to set the coupled item as an unknown estimated parameter in Section 3.

In general, the working-side system is subjected to parametric uncertainties due to the uncertainties of A_1, A_2, C_d, W, ρ, k_{v1}, ω_{v1}, ξ_{v1} and the variations of m, F_f, β_e, V_1, ΔP_1. At the same time, system dynamic is greatly influenced by the load force F and the disturbance Δf, ΔQ_1. To achieve a better dynamic performance and system stability, uncertain parameters must be adaptively adjusted and the disturbance should be rejected while the modeling errors of the flow mapping are compensated. In order to use parameter adaptation to reduce parametric uncertainties for an improved performance, it is necessary to linearly parameterize the state–space equation in terms of a set of unknown parameters. To achieve this, define the unknown parameter as $\theta^w = [\theta_1^w \quad \theta_2^w \quad \theta_3^w \quad \theta_4^w]^{\mathrm{T}} = [\frac{P_{2n}A_2 + F_{fn} + F_n}{m} \quad \frac{\beta_e A_1}{V_{1n}} \quad \omega_{v1}^2 \quad 2\xi_{v1}\omega_{v1}]^{\mathrm{T}}$, define the virtual control gain vector as $b^w = [b_1^w \quad b_2^w \quad b_3^w]^{\mathrm{T}} = [\frac{A_1}{m} \quad \frac{\beta_e C_d W \sqrt{\frac{2}{\rho}\Delta P_{1n}}}{V_{1n}} \quad k_{v1}\omega_{v1}^2]^{\mathrm{T}}$, define the disturbances as $\Delta_2^w = -\frac{\Delta f}{m} + \frac{(P_2 - P_{2n})A_2 + F_f - F_{fn} + F - F_n}{m}$, $\Delta_3^w = -\frac{\Delta Q_1 \beta_e}{V_1} + \beta_e A_1(\frac{1}{V_1} - \frac{1}{V_{1n}})x_2^w + \beta_e C_d W \sqrt{\frac{2}{\rho}}(\frac{\sqrt{\Delta P_1}}{V_1} - \frac{\sqrt{\Delta P_{1n}}}{V_{1n}})x_4^w$, define the control input as $u^w = u_{v1}$, where $\bullet_n$ represents the nominal value of $\bullet$ or the part which can be estimated in adaptive control, while the other part $\bullet - \bullet_n$, which cannot be estimated, is treated as a disturbance and will be rejected by the designed robust controller in the following IARDSC. Then, the state–space (8) can be linearly parameterized as

$$\begin{cases} \dot{x}_1^w = x_2^w \\ \dot{x}_2^w = b_1^w x_3^w - \theta_1^w + \Delta_2^w \\ \dot{x}_3^w = b_2^w x_4^w - \theta_2^w x_2^w + \Delta_3^w \\ \dot{x}_4^w = x_5^w \\ \dot{x}_5^w = b_3^w u^w - \theta_3^w x_4^w - \theta_4^w x_5^w \end{cases} \tag{9}$$

2.2. The off-side system

The objective of the off-side pressure regulator is to control the off-side pressure to reach the desired backpressure P_{0d}. Similarly, take P_2 as an example of the off-side to illustrate the pressure regulator design. Choose state variables as $x^o = [x_1^o \quad x_2^o \quad x_3^o]^{\mathrm{T}} \triangleq [P_2 \quad x_{v2} \quad \dot{x}_{v2}]^{\mathrm{T}}$, the off-side system dynamic state function can be described as

$$\begin{cases} \dot{x}_1^o = \dfrac{\beta_e}{V_2}\left(-C_d W x_2^o \sqrt{\dfrac{2}{\rho}\Delta P_2} + A_2 \dot{x}_L + \Delta Q_2\right) \\ \dot{x}_2^o = x_3^o \\ \dot{x}_3^o = k_{v2}\omega_{v2}^2 u_{v2} - \omega_{v2}^2 x_2^o - 2\xi_{v2}\omega_{v2} x_3^o \end{cases} \tag{10}$$

where $\Delta Q_2 = \Delta Q_{20} + \tilde{Q}_2$, $A_2 \dot{x}_L$ can be seen as the coupled part. For simplicity and decoupling, (10) is simplified to a linearly parameterized state space form with uncertainties.

Similarly, the off-side system is subjected to parametric uncertainties due to the uncertainties of A_2, C_d, W, ρ, k_{v2}, ω_{v2}, ξ_{v2} and the variations of β_e, V_2, ΔP_2. At the same time, system dynamic is greatly influenced by the disturbance ΔQ_2. To adjust uncertain parameters adaptively and reject the disturbance, define the unknown parameter set as $\theta^o = \left[\frac{\beta_e A_2 \dot{x}_{Ln}}{V_{2n}} \quad \omega_{v2}^2 \quad 2\xi_{v2}\omega_{v2}\right]^{\mathrm{T}}$, define the virtual control gain vector as $b^o = \left[\frac{\beta_e C_d W \sqrt{\frac{2}{\rho}\Delta P_{2n}}}{V_{2n}} \quad k_{v2}\omega_{v2}^2\right]^{\mathrm{T}}$, define the disturbance set as $\Delta_2^o = \frac{\Delta Q_2 \beta_e}{V_2} + \beta_e A_2(\frac{\dot{x}_L}{V_2} - \frac{\dot{x}_{Ln}}{V_{2n}}) + \beta_e C_d W \sqrt{\frac{2}{\rho}}(\frac{\sqrt{\Delta P_2}}{V_2} - \frac{\sqrt{\Delta P_{2n}}}{V_{2n}})x_2^o$, and define the control input as $u^o = u_{v2}$. Then, system (10) can be simplified as

$$\begin{cases} \dot{x}_1^o = -b_1^o x_2^o + \theta_1^o + \Delta_2^o \\ \dot{x}_2^o = x_3^o \\ \dot{x}_3^o = b_2^o u^o - \theta_2^o x_2^o - \theta_3^o x_3^o \end{cases} \tag{11}$$

Note that b^w and b^o should be positive definite. Mostly, one of the two systems will transfer to the other with the change of given motion tasks or working modes (Liu & Yao, 2002).

3. Controller design

For simplicity, some notations and assumptions should be made before designing the controller. Define all unknown parameters as a vector $\theta_b = [b^{\mathrm{T}}, \theta^{\mathrm{T}}]^{\mathrm{T}}$. Throughout this paper, the following nomenclature is used: $\tilde{\bullet}$ is used to denote the parameter estimation error of $\bullet$, $\hat{\bullet}$ is used to denote the estimate of $\bullet$, e.g., $\tilde{\theta} = \hat{\theta} - \theta$. $\bullet_{\max}$ and $\bullet_{\min}$ are the maximum and minimum value of $\bullet(t)$ for all t respectively. $\bullet_i$ is the ith component of the vector $\bullet$. Moreover, the following practical assumptions are made:

Assumption 1. The unknown parameter vector θ_b lies within a known closed and bounded convex set Ω, it is assumed that $\forall \theta_b \in \Omega$, $\theta_{i\min} \le \theta_i \le \theta_{i\max}$ and $0 < b_{i\min} \le b_i \le b_{i\max}$, where $\theta_{i\min}$, $\theta_{i\max}$, $b_{i\min}$ and $b_{i\max}$ are some known constants.

Assumption 2. The nonlinear uncertainty Δ is supposed to be bounded by $|\Delta| \le \delta$, where δ is a known positive constant.

Assumption 3. The output desired trajectory x_{Ld} is continuous and feasible, and $[x_{Ld}, \dot{x}_{Ld}, \ddot{x}_{Ld}]^{\mathrm{T}} \in \Omega_{Ld}$, with a known compact set $\Omega_{Ld} = \{[x_{Ld}, \dot{x}_{Ld}, \ddot{x}_{Ld}]^{\mathrm{T}} : x_{Ld}^2 + \dot{x}_{Ld}^2 + \ddot{x}_{Ld}^2 \le M_0\} \subset R^3$ whose size M_0 is a known positive constant.

Due to the appearance of the uncertain nonlinearities Δ in (9) and (11), indirect adaptive robust control method is adopted to achieve the separation of controller and identifier designs. In this section, a type of adaptation law with bounds-varying projection is given out firstly. Then, the IARDSCs for working-side system and the off-side system are designed respectively. Finally, one kind of fast parameter estimation algorithm is proposed to achieve a good tracking performance.

3.1. Bounds-varying projection type adaptation law

In the SMISMO control system, the coupled items between the working-side system and the off-side system are time varying states. To achieve a fast estimation and guarantee a robust performance, a bounds-varying projection mapping is designed in this section. The varying bound indicates a more precise scope of the real parameters rather than the maximum range. With the assumption that the varying bound is a known function of observable states, the new projection mapping can be designed for a fast estimating process by changing the direction of the adaptation law when the estimates exceed the varying bound. The sufficient condition for the convergence of estimation errors is also given out in the following.

Considering a general case and noting Assumption 1, a closed and bounded convex set Ω_θ inside Ω can be preset. Then, the following projection mapping is used to fast guide the estimation value $\hat{\theta}$ towards Ω_θ when $\hat{\theta}$ is out of Ω_θ and then keep the parameter estimates within the bound Ω_θ. The determination of Ω_θ is based on the knowledge of some detectable parameters. To achieve the fast parameter estimation, a new projection type function is proposed:

$$proj_{\hat{\theta}}(\zeta) = \begin{cases} \zeta & \hat{\theta} \in \overset{\circ}{\Omega}_{\theta t} \ or\ n_{\hat{\theta}}^{\mathrm{T}}\zeta \le 0 \\ (I - \Gamma \dfrac{n_{\hat{\theta}} n_{\hat{\theta}}^{\mathrm{T}}}{n_{\hat{\theta}}^{\mathrm{T}} \Gamma n_{\hat{\theta}}})\zeta & \hat{\theta} \in \partial\bar{\Omega}_{\theta\max} \ and\ n_{\hat{\theta}}^{\mathrm{T}}\zeta > 0 \\ \zeta + c\Gamma d & \hat{\theta} \in \breve{\Omega}_{\theta t} \end{cases} \tag{12}$$

where $\zeta \in R^n$ is any n-dimension vector, $\Gamma(t) \in R^{n\times n}$ is any time-varying positive definite symmetric matrix. Denote $\overset{\circ}{\Omega}_{\theta t}$ and $\breve{\Omega}_{\theta t}$ as the interior and the external of the set Ω_θ at time t respectively, they are separated by the time varying boundary $\partial\bar{\Omega}_{\theta t}$. $\partial\bar{\Omega}_{\theta\max}$ denotes the maximum bound of $\Omega_{\theta t}$ independent of time t. In other words, the varying bound $\partial\bar{\Omega}_{\theta t}$ of $\Omega_{\theta t}$ varies with t and the global bound $\partial\bar{\Omega}_{\theta\max}$ is constant. Assume that the mapping between $\partial\bar{\Omega}_{\theta t}$ and observable states can be designed beforehand according to the required performance here. $n_{\hat{\theta}}$ is an outward unit normal vector at $\hat{\theta} \in \partial\bar{\Omega}_\theta$. c is a positive real number related to the estimation speed when parameter changes. $d \in R^n$ is a vector which will be synthesized later. Note that when $c = 0$, (12) becomes the original case described in Krstic et al. (1995) and (Yao & Palmer, 2002).

A special example is shown as follows, in which the dimension of θ is simplified to one.

$$Proj_{\hat{\theta}}(\zeta) = \begin{cases} 0 & \hat{\theta} = \theta_{\max,t}, \zeta > 0 \\ & or\ \hat{\theta} = \theta_{\min,t}, \zeta < 0 \\ a_1\zeta\mathrm{sgn}(\zeta) + \varepsilon & \hat{\theta} < \theta_{\min,t} \\ a_2\zeta\mathrm{sgn}(\zeta) - \varepsilon & \hat{\theta} > \theta_{\max,t} \\ \zeta & \text{otherwise} \end{cases} \tag{13}$$

where $\theta_{\min,t}$ and $\theta_{\max,t}$, related to the off-side pressure P_2 or the load displacement x_L, denote the lower and upper bounds of $\hat{\theta}$ at time t with respect to Ω_θ respectively. Specifically, the mapping between the bounds and states can be prior designed according to the required performance. To ensure the robust performance, a_1 and a_2 satisfy $a_1 > 1$, $a_2 < -1$ and ε is a positive constant.

Lemma 1 (*Hao et al., 2016*). *Suppose that the parameter estimates $\hat{\theta}$ is updated using the following projection type adaptation law:*

$$\dot{\hat{\theta}} = proj_{\hat{\theta}}(\Gamma\tau), \quad \hat{\theta}(0) \in \Omega_\theta \tag{14}$$

where τ is the adaption function and $\Gamma = diag\{\gamma_1, \gamma_2, \dots, \gamma_n\} > 0$ is any continuously differentiable positive symmetric adaptation rate matrix. With this adaption law structure, the following desirable properties hold.

($P1$) *The parameter estimates are always within the known bounded set $\Omega_{\theta t}$ with the known bound $\partial\bar{\Omega}_{\theta\max}$, i.e., $\hat{\theta}(t) \in \Omega_{\theta t}, \forall t$. Thus, from Assumption 1, $\forall t, \theta_{i\min} \le \hat{\theta}_i(t) \le \theta_{i\max}$ and $0 < b_{i\min} \le \hat{b}_i(t) \le b_{i\max}$.*

($P2$)

$$\tilde{\theta}^{\mathrm{T}}(\Gamma^{-1} proj_{\hat{\theta}}(\Gamma\tau) - \tau) \le 0, \quad \forall\tau \tag{15}$$

($P3$) *For time varying parameters, if $\hat{\boldsymbol{\theta}} \in \breve{\Omega}_{\theta t}$, a sufficient condition for the convergence of $\hat{\boldsymbol{\theta}}$ to $\overset{\circ}{\Omega}_{\theta t}$ is shown as:*

$$c > (d^{\mathrm{T}}\Gamma d)^{-1}\sqrt{\|\tau\|^2 - \left(d_\perp^{\mathrm{T}}\tau\right)^2} \tag{16}$$

where $d_\perp$ is unit orthogonal of d.

With the new projection function, Z-swapping identifier introduced in Li et al., (2010) and Yao and Palmer (2002) can be applied in the SMISMO control system. As a result, the proposed bounds-varying projection type on-line adaptation laws can be used to estimate the unknown system parameters.

3.2. IARDSC design

To avoid the 'explosion of terms' problem, ARC backstepping technique is applied in the controller design combining with DSC approach (Li et al., 2010). However, the parameter estimates in ARC are not accurate enough to achieve a high control performance. To solve this problem, ARC is replaced by IARC. Then, IARDSC can be obtained by combining IARC backstepping technique with DSC approach. The stability proof is similar and omitted as well. According to the system model, the IARDSC for the working-side system and the IARDSC for the off-side system should be designed separately. The detailed procedure refers to Chen et al. (2017). If the IARDSC shifts from controlling working-side system to off-side system or the other way round during operations, the improved working mode selection (Liu & Yao, 2002) should been set firstly to avoid cavitation and vibration. Therefore, the proposed IARDSC approach (Chen et al., 2017) is employed to accomplish the two objectives since both parametric uncertainties and nonlinear uncertainties exist in the working-side system (9) and off-side system (11).

Remark 1. For concentrating on energy saving techniques, the detailed IARDSC design procedure, which can be found in Chen et al. (2017), is omitted. As for the state dependent disturbances acting on the system, such as friction depending on displacement and velocity, two methods can be used to deal with the problem. One is model identification in Chen et al. (2015). The other is still taking the state dependent disturbances as normal disturbances, then they can be rejected by the robust item in the IARDSC design (Chen et al., 2017).

3.3. Fast parameter estimation algorithm

In the SMISMO system dynamics (9) and (11), although parameters θ_1^w and θ_1^o will change, this kind of parameter change is generally with a relative large time constant, which means that when the change is very slow, the normal parameter estimation method (Li et al., 2010) can be easily applied in this case. However, it is not sure that the change rates of the two parameters are always under a certain low value. Thus, the parameter estimation rate or the adaptive rate should be always faster than the change rates of parameters. In this circumstance, to protect the mechanical system and maintain a good tracking performance, a fast parameter estimation is needed. For simplicity, the analysis in the rest of this paper is based on the working-side system and the superscript 'w' is neglected.

After the design of IARDSC, a suitable adaptive estimation function for the system needs to be designed. Herein, the system dynamics can be rewritten as

$$\dot{x} = F^{\mathrm{T}}(x,u)\theta_b + f(x) + \Delta \tag{17}$$

where the matrix F is $F^{\mathrm{T}}(x,u) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ x_3 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & x_4 & 0 & 0 & -x_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & u & 0 & 0 & -x_4 & -x_5 \end{bmatrix}$; the vector of known function $f \in R^5$, $f(x) = \begin{bmatrix} x_2 & 0 & 0 & x_5 & 0 \end{bmatrix}^{\mathrm{T}}$, represents the lumped effect of all known nonlinearities and is added for generality; the disturbance vector $\Delta = \begin{bmatrix} 0 & \Delta_2 & \Delta_3 & 0 & 0 \end{bmatrix}^{\mathrm{T}}$.

In order to achieve the fast parameter estimation, a suitable estimation function τ should be constructed so that an asymptotic tracking or zero final tracking error can be obtained in the presence of parametric uncertainties only. To achieve it, the transformed tracking error dynamics is replaced by the original system model (9) to design the parameter estimation (Yao & Palmer, 2002). As such, a more common fast parameter estimation algorithm is proposed based on (Li et al., 2010). Assume the system as a system without uncertain nonlinearities, i.e., let $\Delta_i = 0 (i = 2, 3)$ in system (9), then the system dynamics (9) can be rewritten as

$$\dot{x} = F^{\mathrm{T}}(x,u)\theta_b + f(x) \tag{18}$$

Construct the filters as follow

$$\begin{aligned} \dot{\Omega}^{\mathrm{T}} &= A\Omega^{\mathrm{T}} + F^{\mathrm{T}} \\ \dot{\Omega}_0 &= A(\Omega_0 + x) - f \end{aligned} \tag{19}$$

where A is an exponentially stable matrix. Let $y = x + \Omega_0$ and combining (17), (18) and (19) yields

$$\begin{aligned} \dot{y} &= F^{\mathrm{T}}\theta_b + f + A(\Omega_0 + x) - f \\ &= F^{\mathrm{T}}\theta_b + A(\Omega_0 + x) \end{aligned} \tag{20}$$

If let $\tilde{\varepsilon} = x + \Omega_0 - \Omega^{\mathrm{T}}\theta_b$, which is the filter error, then it is easy to verify that

$$y = \Omega^{\mathrm{T}}\theta_b + \tilde{\varepsilon} \tag{21}$$

where $\tilde{\varepsilon}$ is governed by $\dot{\tilde{\varepsilon}} = A\tilde{\varepsilon}$ and exponentially decays to zero.

Now define the estimate of y and the prediction error ε as

$$\begin{aligned} \hat{y} &= \Omega^{\mathrm{T}}\hat{\theta}_b \\ \varepsilon &= \hat{y} - y = \Omega^{\mathrm{T}}\hat{\theta}_b - x - \Omega_0 \end{aligned} \tag{22}$$

Then, the resulting prediction error model is

$$\varepsilon = \Omega^{\mathrm{T}}\tilde{\theta}_b - \tilde{\varepsilon} \tag{23}$$

Thus, easy to know that the static model (21) is linearly parameterized in terms of $\tilde{\theta}_b$ and with an additional term $\tilde{\varepsilon}$ decaying to zero exponentially.

Define

$$\begin{aligned} M(t) &= \int_0^t \Omega(\varsigma)\Omega^{\mathrm{T}}(\varsigma)d\varsigma \\ N(t) &= \int_0^t \Omega(\varsigma)(y(\varsigma) - \tilde{\varepsilon}(\varsigma))d\varsigma \end{aligned} \tag{24}$$

and from (21), $M\theta_b = N$. Then , choose a novel adaptation law (Li et al., 2010) as

$$\dot{\hat{\theta}}_b = proj_{\hat{\theta}_b}(\Gamma(\lambda(M\hat{\theta}_b - N) - \tau_0)) \tag{25}$$

where λ is a positive learning factor, τ_0 is an initial adaptive function about the error surface (Li et al., 2010) and can be obtained in the IARDSC design procedure (Chen et al., 2017).

4. Energy saving

In this section, two main ways to reduce power consumption in the SMISMO control system were discussed. Traditionally, mobile hydraulic valve and electro-hydraulic load sensing means (Hansen, Pedersen, Andersen, & Wachmann, 2011; Liu, Xu, Yang, & Zeng, 2009d), such as load-sensing variable pump, have been applied in EHSSs. But the energy saving will be analyzed from another perspective, that is using the disturbance observer to control the supply pressure and employing the grey model prediction into the supply flow rate controller.

4.1. Reduce the fluid source supply pressure $P_s(t)$

Generally, the supply pressure is set to be a constant and relative high value to handle all the load conditions. It is one way to guarantee the security, but it is not energy saving when the system is usually running with a small load. Since the load force has a direct relationship with the supply pressure, the supply pressure can be controlled according to the load force. In order to obtain the load force, a force sensor can be mounted between the load and cylinder rod. However, a force sensor is expensive, fragile and hard to mount without any influence on the motion tasks of load. Therefore, a disturbance observer is used here to replace a force sensor to observe the load force.

In the system (2), the force F_f, F, and Δf can be taken as disturbance forces f_d and it can be written as

$$f_d = A_1P_1 - A_2P_2 - m\ddot{x}_L = F_f + F + \Delta f \tag{26}$$

Then, reset the states $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^{\mathrm{T}} \triangleq \begin{bmatrix} x_L & \dot{x}_L \end{bmatrix}^{\mathrm{T}}$, the control input $u = P_1A_1 - P_2A_2$, the control input gain $b = \frac{1}{m}$, the control output $y = x_1$, and the unmeasurable disturbance $\omega = -\frac{f_d}{m}$ in the working-side system (9) to simplify the fifth order system into a second order system yields

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = bu + \omega = \dfrac{1}{m}(u - f_d) \\ y = x_1 \end{cases} \tag{27}$$

Assumption 4. The disturbance ω is continuous and satisfies the following condition

$$\left| \frac{d^j\omega(t)}{dt^j} \right| \le \rho, j = 0, \dots, o \tag{28}$$

where ρ is a positive number, which is not required to know its bound; o represents the order of the disturbance observer to be discussed later and here $o = 2$.

In the system in Fig. 1, a force sensor can be utilized to detect the load, and then work out a proper low P_s to control the relief valve if possible. From another view, due to the relationship between the load and supply pressure, a disturbance observer (Pi & Wang, 2010) can be designed as follows to observe the load to obtain a proper supply pressure, which is low enough.

Firstly, a second order disturbance observer for system (9) is constructed as follows:

$$\begin{aligned} \hat{\omega} &= g_{11} + h_{11}x_2 = g_{11} + h_{11}\dot{x}_L \\ \dot{g}_{11} &= -h_{11}(bu + \hat{\omega}) + \hat{\dot{\omega}} = -h_{11}(\frac{P_1A_1 - P_2A_2}{m} + \hat{\omega}) + \hat{\dot{\omega}} \\ \hat{\dot{\omega}} &= g_{12} + h_{12}x_2 = g_{12} + h_{12}\dot{x}_L \\ \dot{g}_{12} &= -h_{12}(bu + \hat{\omega}) = -h_{12}(\frac{P_1A_1 - P_2A_2}{m} + \hat{\omega}) \end{aligned} \tag{29}$$

where g_{11} and g_{12} are auxiliary variables; h_{11} and h_{12} are the constants chosen by user.

Then, the observer of disturbance force can be obtained by

$$\hat{f}_d = -m\hat{\omega} \tag{30}$$

Define the estimation errors as

$$\tilde{e} = \begin{bmatrix} \tilde{\omega} & \tilde{\dot{\omega}} \end{bmatrix}^{\mathrm{T}} \triangleq \begin{bmatrix} \hat{\omega} - \omega & \hat{\dot{\omega}} - \dot{\omega} \end{bmatrix}^{\mathrm{T}} \tag{31}$$

From (27) and (29)

$$\dot{\hat{\omega}} = \dot{g}_{11} + h_{11}\dot{x}_2 = -h_{11}\tilde{\omega} + \hat{\dot{\omega}} \tag{32}$$

Let both sides of (32) be subtracted by $\dot{\omega}$ yields

$$\dot{\tilde{\omega}} = \dot{\hat{\omega}} - \dot{\omega} = -h_{11}\tilde{\omega} + \hat{\dot{\omega}} - \dot{\omega} = -h_{11}\tilde{\omega} + \tilde{\dot{\omega}} \tag{33}$$

Similarly, from (27) and (29),

$$\dot{\tilde{\dot{\omega}}} = \dot{\hat{\dot{\omega}}} - \ddot{\omega} = \dot{g}_{12} + h_{12}\dot{x}_2 - \ddot{\omega} = -h_{12}\tilde{\omega} - \ddot{\omega} \tag{34}$$

Noting (34) and differentiating (33)

$$\ddot{\tilde{\omega}} = -h_{11}\dot{\tilde{\omega}} + \dot{\tilde{\dot{\omega}}} = -h_{11}\dot{\tilde{\omega}} - h_{12}\tilde{\omega} - \ddot{\omega} \tag{35}$$

Since $\ddot{\omega}$ is bounded as Assumption 4, for stability of $\tilde{\omega}$, it is necessary and sufficient to select $h_{11} > 0$ and $h_{12} > 0$. Noting that the disturbance observer estimates ω as well as $\dot{\omega}$. The observer error dynamics in compact form can be written as

$$\begin{aligned} \dot{\tilde{e}} &= C\tilde{e} + D\ddot{\omega} \\ C &= \begin{bmatrix} -h_{11} & 1 \\ -h_{12} & 0 \end{bmatrix}, D = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \end{aligned} \tag{36}$$

The analysis of the accuracy of estimation and the stability of the disturbance observer are discussed in Appendix.

Thus, the supply pressure can be given by

$$P_s = k_f \left| \hat{f}_d \right| + k_v \left| \dot{x}_L \right| \tag{37}$$

where k_f and k_v represent the pressure coefficients of load disturbance and velocity respectively. (37) shows that the supply pressure is related to the disturbance and the motion of load. In the context of system stability and cavitation prevention, from (29), reducing P_{0d} can also reduce the fluid source supply pressure P_s.

From (37), a low enough supply pressure P_s can be set to save energy by choosing the proper k_f and k_v. However, supply pressure P_s cannot be too low to handle the load force and motion. Define ΔW_1 and ΔW_2 as the throttling energy loss of PDCV1 and PDCV2 respectively as follows

$$\begin{aligned}\Delta W_1 &= \dot{x}_L A_1(P_s - P_1)\\ \Delta W_2 &= \dot{x}_L A_2(P_2 - P_r)\end{aligned} \tag{38}$$

Then, the sum of throttling energy loss ΔW is

$$\Delta W = \Delta W_1 + \Delta W_2 = \dot{x}_L A_2(\frac{A_1}{A_2}P_s - P_r - \frac{P_L}{A_2}) \tag{39}$$

Set

$$\frac{\partial \Delta W}{\partial \dot{x}_L} = 0 \Rightarrow A_2(\frac{A_1}{A_2}P_s - P_r - \frac{P_L}{A_2}) = 0 \tag{40}$$

to obtain the minimum of P_s

$$P_{s\,\min} = \frac{A_2}{A_1}P_r + \frac{P_L}{A_1} + \Delta P_s \tag{41}$$

where ΔP_s is the extra margin of the supply pressure, which is set to a small value to maintain the enough driving force and safety motion of load. Thus, the choices of k_f and k_v in (37) are constrained by (41). Based on (37), the related control voltage input of the relief valve (Chen et al., 2015) can be obtained.

4.2. Reduce the fluid source flow rate $Q_s(t)$

As shown in Fig. 1, the load-sensing variable pump, implemented through a fixed displacement pump derived by a servo motor, is a novel scheme. If the demand of flow rate is known, the related rotate speed n can be obtained to control the servo motor. Thus, a flow rate predictor is needed.

Grey predictor is one kind of predictor (Julong, 1989), which can be utilized to predict the future system outputs with high accuracy, even if the mathematical model of the real system is unknown. GM(m, n) can be used to denote a grey model in grey prediction theory, where n is the order of the difference equation and m is the number of variables. The grey predictor can conduct an accumulated generating operation (AGO) on an original sequence. By using the least-square method, the resultant series can be utilized to establish a difference equation to calculate coefficients. Then, the accumulated generating series of the prediction model can be obtained. Through inverse accumulated generating operation (IAGO), the value can be taken to estimate the future output of system in the time-domain. Therefore, one of the most popular grey models, GM(1, 1) (Grey Model First Order One Variable) (Zeng et al., 2010), can be used to predict the displacement purposes in this research.

Firstly, at least four output data points are needed to approximate the system (9). For a nonnegative time series, r rows data is sampled as:

$$\begin{aligned}x_L^{(0)} &= \{x_L^{(0)}(1), x_L^{(0)}(2), \ldots, x_L^{(0)}(r)\}\\ x_L^{(0)}(j) &\geq 0, j = 1, 2, \ldots, r\end{aligned} \tag{42}$$

Use the AGO (Chiang & Tseng, 2004) to obtain $x_L^{(1)}$ from $x_L^{(0)}$

$$x_L^{(1)}(j) = \sum_{i=1}^{j} x_L^{(0)}(i), j = 1, 2, \ldots, r \tag{43}$$

Noting that the superscripts '(0)' and '(1)' do not denote the time derivative only in this subsection.

By using the mean generating operation (MGO) as follows, a consecutive neighbor generation $y^{(1)}$ can be obtained from $x_L^{(1)}$

$$y^{(1)}(j) = 0.5x_L^{(1)}(j) + 0.5x_L^{(1)}(j-1), j = 2, 3, \ldots, r \tag{44}$$

Then, a grey differential equation of GM(1, 1) is established

$$x_L^{(0)}(j) + ay^{(1)}(j) = b \tag{45}$$

where a and b are fitting parameters, which can be obtained via the following least square method

$$K = \begin{bmatrix} a\\ b\end{bmatrix} = \left(J^{\mathrm{T}}J\right)^{-1}J^{\mathrm{T}}Y \tag{46}$$

where

$$J = \begin{bmatrix} -y^{(1)}(2) & 1\\ -y^{(1)}(3) & 1\\ \vdots & \vdots\\ -y^{(1)}(r) & 1\end{bmatrix}, Y = \begin{bmatrix} x_L^{(0)}(2)\\ x_L^{(0)}(3)\\ \vdots\\ x_L^{(0)}(r)\end{bmatrix} \tag{47}$$

And next, the prediction model GM(1, 1) can be set up as

$$\begin{aligned}\hat{x}_L^{(1)}(j+1) &= \left(\hat{x}_L^{(1)}(1) - \frac{b}{a}\right)e^{-aj} + \frac{b}{a}\\ \hat{x}_L^{(0)}(j+1) &= \hat{x}_L^{(1)}(j+1) - \hat{x}_L^{(1)}(j)\end{aligned} \tag{48}$$

Finally, the predictive output at time sequence $(k+v)$th step can be calculated by

$$\begin{aligned}\hat{x}_L^{(1)}(k+v) &= \left(\hat{x}_L^{(1)}(1) - \frac{b}{a}\right)e^{-a(k+v-1)} + \frac{b}{a}\\ \hat{x}_L^{(0)}(k+v) &= \hat{x}_L^{(1)}(k+v) - \hat{x}_L^{(1)}(k+v-1)\end{aligned} \tag{49}$$

where k is the step size of the grey prediction and $k+v$ means the displacement of load can be obtained before v control cycles. Thus, the influence of the delay and drop of flow rate and pressure caused by the length of pipeline from the pump station to the valve can be eliminated by choosing a proper v.

Also, the total flow rate of system can be derived

$$Q_S = Q_L + Q_C \tag{50}$$

where Q_C is the least leak flow rate of relief valve or a flow margin (the pump must be providing greater than the required flow) required to ensure the relief valve is always open, and

$$Q_L(k) = A\dot{\hat{x}}_L^{(0)}(k+v), A = \begin{cases} A_1 & \dot{\hat{x}}^{(0)}(k+v) > 0\\ 0 & \dot{\hat{x}}^{(0)}(k+v) = 0\\ A_2 & \dot{\hat{x}}^{(0)}(k+v) < 0\end{cases} \tag{51}$$

Thus, the desired rotate speed of motor can be obtained by

$$n = \begin{cases} \dfrac{Q_S + C_P P_s}{D} + \Delta n & if\ \dfrac{Q_S + C_P P_s}{D} > n_0 - \Delta n\\ n_0 & if\ \dfrac{Q_S + C_P P_s}{D} \leq n_0 - \Delta n\end{cases} \tag{52}$$

where C_P is the leakage coefficient of pump, D is the displacement of pump, n_0 represents the lowest operating rotate speed of servo motor, and Δn is an extra rotate speed of servo motor, which is utilized to reject the disturbances acting on the servo motor.

5. Experimental results

In order to validate the feasibility and effectiveness of the proposed IARDSC, a self-made SMISMO control system platform was set up firstly. Then, the proposed controller was successfully implemented to control the SMISMO control system and its control performance was compared with PI, ARC&IARC in the conventional electro-hydraulic servo system (CEHSS). Also, the similar experimental results, omitted here, can be obtained in the SMISMO control system as well. Then, some intermediate variables of system including pressures, flow rates, results of disturbance observer and grey predictor, rotate speeds, and control inputs were discussed for why energy saving is achievable in the SMISMO control system. Finally, the proposed two energy saving techniques were employed in the SMISMO control system and analyzed

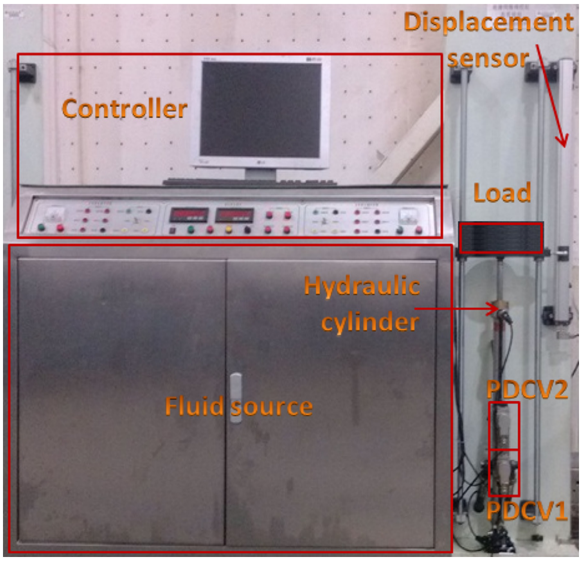

Fig. 2. The constructed SMISMO control system.

by the comparative experimental results between SMISMO control system and CEHSS.

Assumed the offside pressure as a constant value to simplify the design procedure of IARDSC (Liu & Yao, 2002) and employed the designed IARDSC in experiments. For highlighting the significant performance the SMISMO control system can achieve in the field of power consumption and efficiency, a sine motion trajectory tracking was analyzed in this section and the given trajectory was $x_{Ld} = 0.25 + 0.25\sin(\frac{\pi t}{2} - \frac{\pi}{2})$ m.

5.1. The performance of IARDSC

The special SMISMO control system prototype was constructed with an inertia load. The photo of the prototype is shown in Fig. 2, which is based on the experimental schematic setup of the designed SMISMO control system as shown in Fig. 1. The constructed SMISMO control system mainly consists of five parts: the inertia load, the fluid source, the controller, the hydraulic cylinder, and two PDCVs. Noting that the fluid source includes a relief value to control the supply pressure and a servo motor to control the supply flow rate.

As shown in Fig. 1, there were four control inputs: u_{v1}, u_{v2}, P_s, n and six outputs: x_L, f_d, $\dot{x}_L$, P_1, P_2, P_s. Actually, the fluid source supply pressure was implemented by relief valve pressure servo system and the fluid source supply flow rate was implemented by motor rotate speed servo system.

Table 1 shows some of system parameters and key components manufacturer/model. Through a tedious unit conversion and calculation, the maximum and minimum values of estimated parameters can be set as:

$\theta^w_{\min} = [1, 2.06\times10^4, 500, 57]^{\mathrm{T}}$,
$\theta^w_{\max} = [35, 5.28\times10^4, 1\times10^4, 256]^{\mathrm{T}}$,
$b^w_{\min} = [4.91\times10^{-6}, 5.3\times10^7, 49]^{\mathrm{T}}$,
$b^w_{\max} = [5\times10^{-4}, 3\times10^8, 128]^{\mathrm{T}}$,
$\theta^o_{\min} = [1328, 676, 49]^{\mathrm{T}}$,
$\theta^o_{\max} = [1.3\times10^4, 1.45\times10^4, 256]^{\mathrm{T}}$,
$b^o_{\min} = [3.9\times10^7, 25]^{\mathrm{T}}$,
$b^o_{\max} = [2.7\times10^8, 196]^{\mathrm{T}}$.

In experiments, the time-varying bounds $\partial\bar{\Omega}_{\theta t}$ of θ^w_1 and θ^o_1 were chosen as
$[\frac{(P_{0d}-\Delta P_{0d})A_2+F_f+F}{m}, \frac{(P_{0d}+\Delta P_{0d})A_2+F_f+F}{m}]$
and
$[\frac{\beta_e A_2(x_{Ld}-\Delta x_{Ld})}{V_2}, \frac{\beta_e A_2(x_{Ld}+\Delta x_{Ld})}{V_2}]$
respectively, where ΔP_{0d} and Δx_{Ld} are respectively the perturbation margins of desired backpressure and given motion trajectory. Moreover,

Table 1
System parameters and components manufacturer/model.

P_s	60 bar	P_r	0 bar
A_1	4.91×10^{-4} m^2	A_2	2.9×10^{-4} m^2
V_{10}	1.0×10^{-5} m^3	V_{20}	1.0×10^{-5} m^3
P_{0d}	20 bar	C_d	0.62
W	0.0314 m	ρ	870 kg/m^3
β_e	890 MPa	m	50 kg
g	9.8 m/s^2	PDCV	Parker
Hydraulic cylinder	Hanchen	Displacement sensor	Balluff

the other parameters were estimated with the given constant bounds as above, which can be treated as a special case of time-varying bounds. The parameters used in the estimation were chosen as
$A^w = -13\times diag\{0.8, 0.02, 157, 336, 42\}$,
$A^o = -10\times diag\{0.04, 178, 409, 39\}$,
$\Gamma^w = 2.5\times diag\{27, 100, 5, 330, 68, 5500, 940\}$,
$\Gamma^o = 6\times diag\{4, 78, 800, 7300, 560\}$,
$\lambda^w = 0.41$,
$\lambda^o = 0.59$.
The parameters used in the projection were chosen as $a_1 = 1.17$ and $a_2 = -1.25$. Here, set $\Delta P_{0d} = 5$ bar, $\Delta x_{Ld} = 0.025$ m, then the time varying bound $\partial\bar{\Omega}_{\theta t}$ of θ^w_1 and θ^o were
$[5.8\times10^{-6}P_{0d} + 8.7, 5.8\times10^{-6}P_{0d} + 14.5]$
and
$[25810x_{Ld} - 645.25, 25810x_{Ld} + 645.25]$
respectively. The constant load force was $F = mg$. The friction force F_f can be obtained through Stribeck Friction Model identification (Chen et al., 2015) as

$$F_f(\dot{x}_L) = \begin{cases} 68 + 13\dot{x}_L + 11e^{[-3(\dot{x}_L/0.5)]}, \dot{x}_L > 0 \\ -79 + 24\dot{x}_L - 16e^{[3(\dot{x}_L/0.6)]}, \dot{x}_L < 0 \end{cases}.$$

The tracking performance of IARDSC was evaluated by employing it in the CEHSS and SMISMO control system, and comparing with the other three controllers: PI, ARC&IARC, in the CEHSS. As shown in Fig. 3, the tracking error of the conventional PI controller in the CEHSS, ARC in the CEHSS, IARC in the CEHSS, IARDSC in the CEHSS, and IARDSC in the SMISMO control system are about 2 mm, 1 mm, 1 mm, 1.7 mm, 1.4 mm, respectively, see Table 2. In addition, the oscillations during the first second in the following figures were because all the experimental data were sampled when the system started up and the system states needed time to reach their required values, such as the supply pressure. Compared with the conventional PI controller in the CEHSS, the better tracking precision of ARC&IARC is because ARC&IARC can adapt the system parameters and reject the internal uncertainties and external disturbances well. Comparing the tracking error of ARC&IARC with IARDSC in the CEHSS, the worse tracking accuracy of IARDSC indicates the first order filters in IARDSC will make the tracking error larger. Besides, the tracking error of IARDSC in the SMISMO control system is smaller than that in the CEHSS. Since there is only one proportional directional control valve in the CEHSS, the existence of the mechanical linkage between the meter-in and meter-out orifices is hard to eliminate. As a result, this kind of system cannot be controlled better and more freely. Therefore, the tracking performance can be improved by using the SMISMO control system, but the first order filters in IARDSC are not good for the tracking performance. However, most importantly, all the controllers have close tracking performances and the tracking errors are within the same magnitude. That reveals that the SMISMO control system is conductive to the control performance, but the first order filters in IARDSC are not. Moreover, the blue dot line in the first plot is the prediction value of $\hat{x}_L$, which is predicted by GM(1, 1).

Because of the multiple differential of the virtual controller, ARC/IARC suffers from high-order disturbance differentiations, making the control input too huge and with a lot of high order disturbances, which cannot be implemented as a real controller. Thus, the ARC/IARC controller is often simplified through ignoring the high order items which are little to make it realizable (Guan & Pan, 2008; Yao et

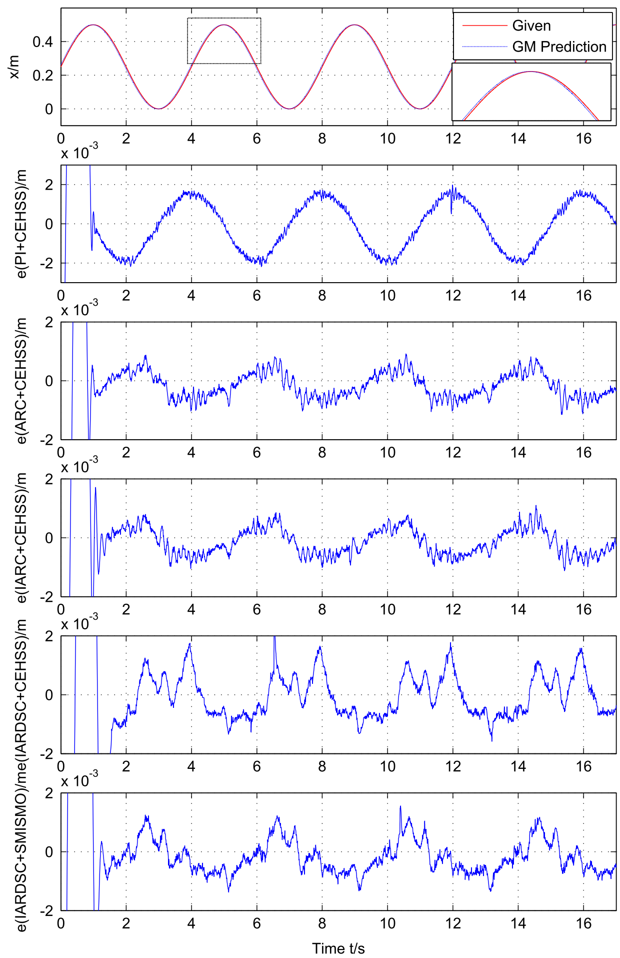

Fig. 3. The tracking performance in experiments.

Table 2
The tracking errors in experiments.

Controller + system	Tracking error
PI + CEHSS	2 mm
ARC + CEHSS	1 mm
IARC + CEHSS	1 mm
IARDSC + CEHSS	1.7 mm
IARDSC + SMISMO	1.4 mm

al., 2000, 2001). Furthermore, low pass filters (<100 Hz) are used to filter out high frequency control inputs. By contrast, the IARDSC is easy to implement without any omission. The high order disturbance differentiations are replaced by the first order filters, which make the IARDSC control input smoother. As a result, the tracking error of IARDSC is smoother than that of ARC&IARC. Therefore, the IARDSC is easy to be utilized in real systems and the experimental results verify that the proposed method is feasible and effective.

As a whole, experiments show that the proposed IARDSC with fast parameter estimation method has a good enough parameter estimation performance to get a good trajectory tracking performance. In addition, the control input and computation cost are both reduced by the introduction of the first order filters when compared with ARC&IARC.

5.2. Energy saving

Generally, too much high fluid source supply pressure and flow rate are not necessary in the EHSS. The extra pressure and flow rate overflowing from the relief valve lead to a lot of energy waste. Thus, two techniques were investigated to save the power consumption: reducing the supply pressure by using a disturbance observer and reducing the supply flow rate by using grey model prediction.

Moreover, in order to further evaluate the energy saving performance of the SMISMO control system and the proposed IARDSC, a set of experiments were carried out, including the CEHSS with IARDSC, the SMISMO control system with IARDSC, and the SMISMO control system with IARDSC and two energy saving techniques (SMISMO + ES). In addition, an external disturbance force (500 N) was added on the load at 9 s, which was represented by a vertical red line in the following figures.

In the experiments of reducing the supply pressure, the parameters of the observer were chosen as: $h_{11} = 110$, $h_{12} = 25$; the pressure coefficient of load disturbance $k_f = 0.0048$ MPa/N and pressure coefficient of velocity $k_v = 1.3$ MPa/(m/s). The first column in Fig. 4 shows the actual pressures P_s, P_1, and P_2 in the sine motion trajectory tracking experiments. From this figure, easy to see that the supply pressure P_s can be reduced to some extent to improve the energy efficiency. Comparing the first and second plot in the first column in Fig. 4, the SMISMO control system makes the control of backpressure possible. Once the off-side pressure is controlled to be low enough, the working-side pressure will not be too high. Then the supply pressure can be reduced to some extent without losing the basic pressure drop in the PDCV. This is how to save energy by reducing supply pressure. The supply pressure is determined by (37), and the minimal supply pressure is based on (41). The two equation indicate that the supply pressure is related to the load force and velocity. The velocity of load can be obtained via the differentiation of the measured displacement of load (the second plot in Fig. 5), which was equipped with a R-C low-pass filter to filter the high frequency noise for the feedback variable. In practical projects, a force sensor is not cheap and its use is limited by the operation space. Thus, a second order disturbance observer was utilized to observe the load force (the first plot in Fig. 5). Actually, from the expression of the disturbance observer (29), it can be found that the load force is worked out through the two chamber pressure. This is a kind of soft measurement principle. Easy to see that the observed load force is almost close to the measured one. As such, the proposed disturbance observer really works. Besides, the estimation accuracy of this kind of observer can be improved with higher order (Ginoya et al., 2014; Yang et al., 2013). From the first and second plot in Fig. 5 and combining (37) and (41), the energy saving performance of the SMISMO control system with IARDSC and the technique of reducing the supply pressure can be addressed. As shown in the third plot in the first column in Fig. 4, the supply pressure P_s is reduced about 50%. which shows the power consumption can be saved about a half in this case. Moreover, after the external disturbance force was added, the pressure of working-side became larger to keep the load going up and became smaller to maintain the downward motion of load (over-running), while the pressure of off-side still could be controlled around the setting value $P_{0d} = 20$ bar. Also the adding external disturbance force was well estimated by the proposed disturbance observer (the first plot in Fig. 5).

In the experiments of reducing the supply flow rate, the lowest operating rotate speed of servo motor was set as $n_0 = 400$ r/min, which was utilized to keep the fundamental supply pressure and flow rate, while the normal operating rotate speed of servo motor was set as $n = 1000$ r/min. And, the extra rotate speed of servo motor was set as $\Delta n = 20$ r/min. The data sampling time of the grey model predictor is the same as one system control cycle time $t = 1$ ms. The number of data rows was chosen as $r = 15$, and the advanced control cycles was given as $v = 30$. The leakage coefficient of pump is about $C_P = 0.02$ mL/s/MPa, and the rated displacement of pump is $D = 12.5$ mL/r. The second column in Fig. 4 shows the actual pressures Q_s, Q_1, and Q_2 in experiments. Comparing the three plots, easy to know that Q_1 and Q_2 are similar and cannot be controlled in the set of experiments. That is because they are determined by the differentiation of given motion trajectory x_{Ld}. Thus, what should be done is only reducing the supply flow rate Q_s here. This

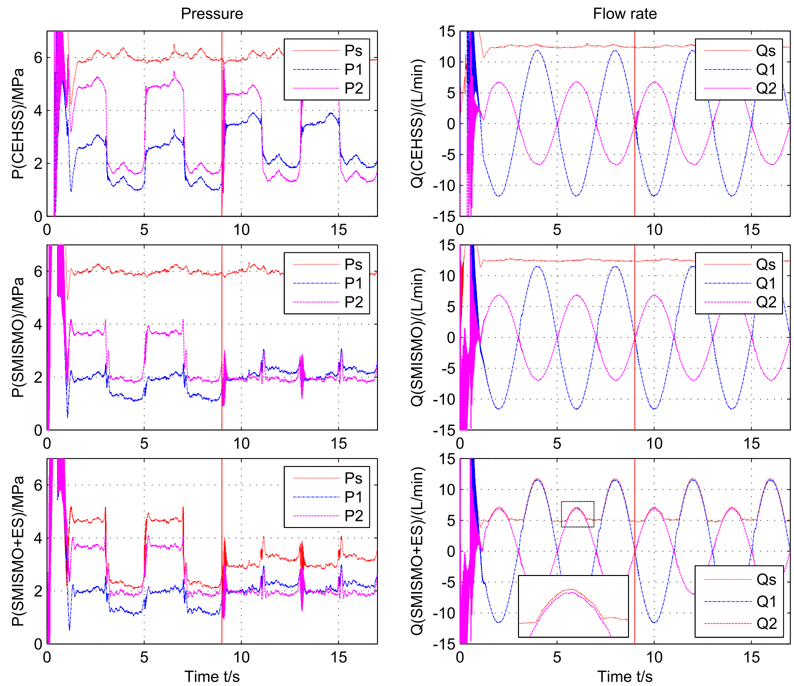

Fig. 4. The actual P_s, P_1, P_2, Q_s, Q_1, Q_2 in experiments.

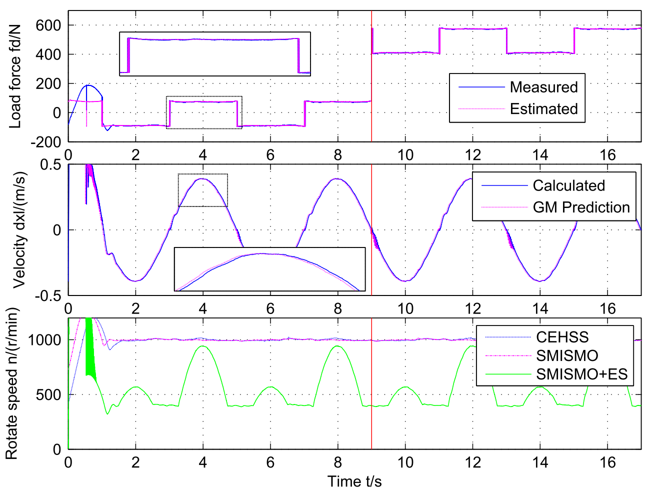

Fig. 5. The actual f_d, $\dot{x}_L$, n in experiments.

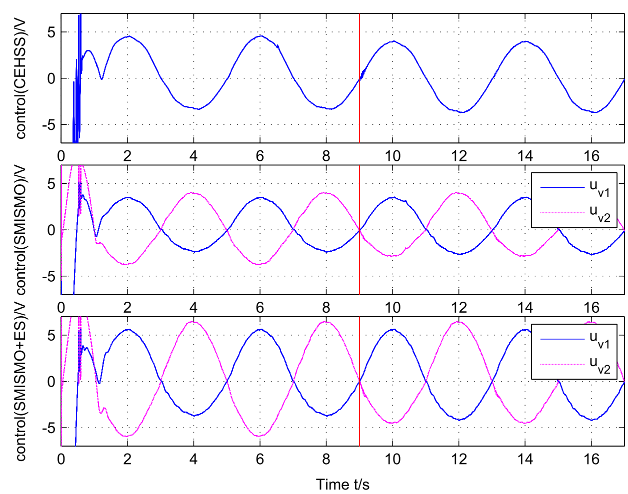

Fig. 6. The corresponding control input in experiments.

implies the supply flow rate should be made as small as possible without losing the basic flow rate difference to keep the supply pressure. Owing to the delay and drop of flow rate and pressure leaded by the length of pipeline from the pump station to the valve, a GM(1,1) is utilized to predict the velocity of the load (the second plot in Fig. 5). Then the demand supply flow rate is obtained through combining the calculated motor rotate speed n (the third plot in Fig. 5 by (52)) and (50). As shown in the third plot in the second column in Fig. 4, the supply flow rate Q_s is reduced about 40%, which is also the percentage of power consumption that can be saved. Therefore, the performance of the SMISMO control system with IARDSC and the technique of reducing the supply flow rate is presented.

The corresponding control inputs in Fig. 4 are shown in Fig. 6. By comparison, the control input of IARDSC in the CEHSS is about 4.7 V, which is larger than that in the SMISMO control system in the working-side (the part below 0 in the second plot). As shown in the second plot in the first column in Fig. 4, the difference between the supply pressure and the pressure of working-side in the SMISMO control system is larger than that in the CEHSS since the pressure of off-side could be controlled to a low value in the SMISMO control system. According to (5), the orifice opening should be smaller in the SMISMO control system to keep the same flow rate maintaining the same motion trajectory. Similarly, the supply pressure in the SMISMO + ES could be controlled to a smaller value comparing with the SMISMO control system, which results in the smaller difference between the supply pressure and the pressure of working-side in the SMISMO + ES. As thus, the orifice opening should be larger based on (5) (the third plot). The larger control input of PDCV represents the larger valve spool opening and the lower throttling loss. The change of control input after adding the external disturbance force can be analyzed in the same way.

To make the energy saving performance of the set of experiments clearer, the power consumptions (P ($P(t) = Q_s(t)P_s(t)$) changing with respect to time) without energy saving techniques (without ES), only with the supply pressure control (only PC), only with the supply flow

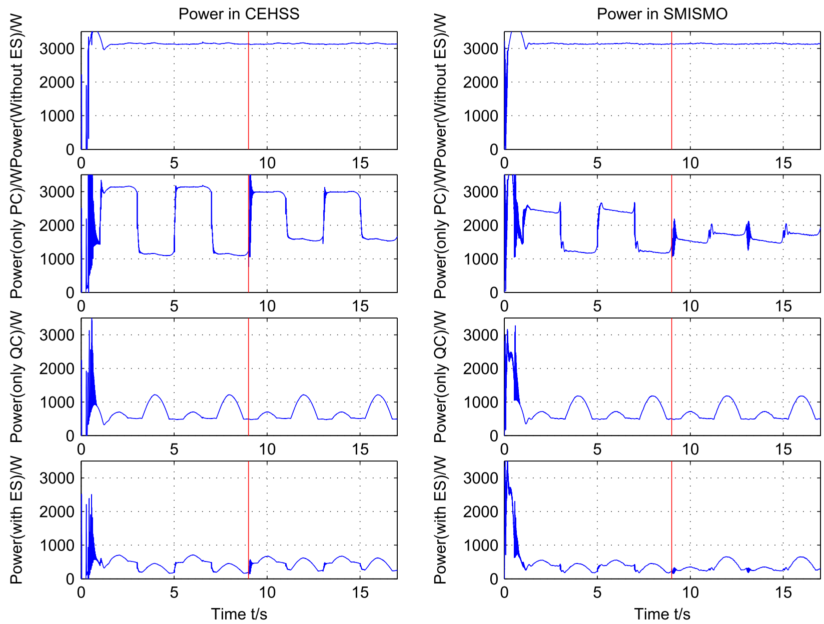

Fig. 7. The actual power consumption P in experiments.

rate control (only QC), with energy saving techniques (with ES) in both CEHSS and SMISMO control system were addressed and shown in Fig. 7. On one hand, no matter in which system, compared with the power consumption without energy saving techniques, the system only with the supply pressure control could save about 1/3 power, the system only with the supply flow rate control could save more than 2/3 power, and the system with energy saving techniques could save about 5/6 power. On the other hand, only with the supply pressure control, the SMISMO control system could save more power than CEHSS since the pressure of off-side in the SMISMO control system can be control to a low level. However, no significant difference in power consumption could be found between CEHSS and the SMISMO control system only with the supply flow rate control since the motions of load were the same. Therefore, the SMISMO control system with energy saving techniques can achieve the best energy saving performance. In addition, the change of power consumption caused by the adding external disturbance force mainly results from the change of pressure of working-side caused by the adding external disturbance force (the third plot in the first column in Fig. 4).

All in all, the SMISMO control system, the proposed IARDSC, and the energy saving techniques by reducing the supply pressure and supply flow rate achieve jointly a good performance of tracking and energy saving.

6. Conclusion

In this paper, a SMISMO control system was considered. To address the internal parameter uncertainties and external disturbances, as well as to solve the 'explosion of terms' and simplify the controller design procedure, an indirect adaptive robust dynamic surface controller was proposed. On the other hand, a fast parameter estimation scheme was utilized to achieve a better estimation performance because of the coupled parts between the working-side system and off-side system. Besides, two classical measures for energy saving in the SMISMO control system: reducing the supply pressure by using a disturbance observer and reducing the supply flow rate by using a grey model predictor, were analyzed and employed. Finally, experimental results showed the effectiveness of the proposed controller with fast estimation scheme, and the performance of energy saving was investigated.

Appendix A. Proof of the stability of disturbance observer

To prove the stability of disturbance observer, a more generalized system with nth order as follow is considered

$$\begin{cases} \dot{x}_1 = x_2 + \omega_1 \\ \dot{x}_2 = x_3 + \omega_2 \\ \vdots \\ \dot{x}_{n-1} = x_n + \omega_{n-1} \\ \dot{x}_n = a(x) + b(x)u + \omega_n \\ y = x_1 \end{cases} \tag{A.1}$$

where the state vector is $x = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^{\mathrm{T}} \in R^n$; $u \in R$ is input signal; $y \in R$ is output signal; $a(x)$ and $b(x)$ are smooth nominal functions. Obviously, there are disturbances in all channels in the system. The unmatched disturbances are $\omega_i (i = 1, \dots, n-1)$, while the matched disturbance is ω_n. Noting that the disturbances may include state dependent and/or external unmeasurable disturbances, nonlinearities and uncertainties.

Assumption 5. The disturbances ω_i are continuous and satisfy the following condition

$$\left| \frac{d^j \omega_i(t)}{dt^j} \right| \leq \rho_i, i = 0, \dots, n; j = 0, \dots, o \tag{A.2}$$

In order to estimate the disturbance $\omega_i (i = 1, \dots, n-1)$ and its derivatives in the ith channel, define

$$\hat{\omega}_i^{(j-1)} = g_{ij} + h_{ij} x_i, i = 0, \dots, n-1 \tag{A.3}$$

where the auxiliary variables g_{ij} are defined as

$$\begin{aligned} \dot{g}_{ij} &= -h_{ij}(x_{i+1} + \hat{\omega}_i) + \hat{\omega}_i^{(j)}, j = 1, \dots, o-1 \\ \dot{g}_{io} &= -h_{io}(x_{i+1} + \hat{\omega}_i) \end{aligned} \tag{A.4}$$

Similarly, the disturbance ω_n and its derivatives can be estimated as

$$\hat{\omega}_n^{(j-1)} = g_{nj} + h_{nj} x_n \tag{A.5}$$

where auxiliary variables g_{nj} are defined as

$$\begin{aligned} \dot{g}_{nj} &= -h_{nj}(a(x) + b(x)u + \hat{\omega}_n) + \hat{\omega}_n^{(j)}, j = 1, \dots, o-1 \\ \dot{g}_{no} &= -h_{no}(a(x) + b(x)u + \hat{\omega}_n) \end{aligned} \tag{A.6}$$

The proper gains h_{ij} of the disturbance observers for the disturbances ω_i can always be given so that all the eigenvalues of each C_i are in the Left Half Plane (LHP). Based on (36), the observer error dynamics can be written in compact form

$$\dot{\tilde{e}}_i = C_i\tilde{e}_i + D_i\omega_i^{(o)}$$
$$C_i = \begin{bmatrix} -h_{i1} & 1 & 0 & \cdots & 0 \\ -h_{i2} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -h_{i(o-1)} & 0 & 0 & \cdots & 1 \\ -h_{io} & 0 & 0 & \cdots & 0 \end{bmatrix}, D_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ -1 \end{bmatrix} \tag{A.7}$$

where $\dot{\tilde{e}}_i = \begin{bmatrix} \tilde{\omega}_i & \tilde{\dot{\omega}}_i & \cdots & \tilde{\omega}_i^o \end{bmatrix}^{\mathrm{T}} (i = 1, \dots, n)$.

Thus, for a given positive definite matrix H_i, finding a positive definite matrix G_i is always possible such that

$$C_i^{\mathrm{T}} G_i + G_i C_i = -H_i \tag{A.8}$$

Define $\lambda_{i\,\min}$ as the smallest eigenvalue of H_i and choose a Lyapunov function as follow

$$V(\tilde{e}_1, \tilde{e}_2, \dots \tilde{e}_n) = \sum_{i=1}^{n} \tilde{e}_i^{\mathrm{T}} G_i \tilde{e}_i \tag{A.9}$$

Then, evaluate $\dot{V}(\tilde{e}_1, \tilde{e}_2, \dots \tilde{e}_n)$ with (A.7)

$$\begin{aligned} \dot{V}(\tilde{e}_1, \tilde{e}_2, \dots \tilde{e}_n) &= \sum_{i=1}^{n} \left(\tilde{e}_i^{\mathrm{T}} (C_i^{\mathrm{T}} G_i + G_i C_i)\tilde{e}_i + 2\tilde{e}_i^{\mathrm{T}} G_i D_i \omega_i^{(o)} \right) \\ &\le \sum_{i=1}^{n} (-\tilde{e}_i^{\mathrm{T}} H_i \tilde{e}_i + 2\left\|\tilde{e}_i\right\| \left\|G_i D_i\right\| \rho_i) \\ &\le \sum_{i=1}^{n} (-\lambda_{i\,\min} \left\|\tilde{e}_i\right\|^2 + 2\left\|\tilde{e}_i\right\| \left\|G_i D_i\right\| \rho_i) \\ &\le -\sum_{i=1}^{n} (\lambda_{i\,\min} \left\|\tilde{e}_i\right\| - 2\left\|G_i D_i\right\| \rho_i) \left\|\tilde{e}_i\right\| \end{aligned} \tag{A.10}$$

From (A.10), the norm of the estimation error will be bounded as follow after a sufficiently long time

$$\left\|\tilde{e}_i\right\| \le \lambda_{i0} = \frac{2\left\|G_i D_i\right\| \rho_i}{\lambda_{i\,\min}} \tag{A.11}$$

Define $\lambda_0 = \max\{\lambda_{i0}\}(i = 1, \dots, n)$, then (A.11) can be rewritten as $\left\|\tilde{e}_i\right\| \le \lambda_0 (i = 1, \dots, n)$.

References

Aardema, J. A. (1996). Hydraulic circuit having dual electrohydraulic control valves, Google patents. US Patent 5,568,759.

Aardema, J. A., & Koehler, D. W. (1999). System and method for controlling an independent metering valve, Google patents, US Patent 5,960,695.

Book, R., & Goering, C. E. (1999). *Programmable electrohydraulic valve, Technical report, SAE technical paper*.

Breeden, R. H. (1981). *Development of a high pressure load sensing mobile valve, Technical report, SAE technical paper*.

Chen, W.-H. (2003). Nonlinear disturbance observer-enhanced dynamic inversion control of missiles. *Journal of Guidance, Control, and Dynamics*, *26*(1), 161–166.

Chen, G., Wang, J., Ma, L., & Hao, R. (2015). Observer-based and energy saving control of single-rod electro-hydraulic servo system driven by servo motor. In *2015 american control conference* (pp. 2224–2229). IEEE.

Chen, G., Wang, J., Wang, S., & Ma, L. (2016). Separate meter in and separate meter out energy saving control system using dual servo valves under complex load conditions. *Transactions of Beijing Institute of Technology*, *36*(10), 1053–1058.

Chen, G., Wang, J., Wang, S., Zhao, J., & Shen, W. (2017). Indirect adaptive robust dynamic surface control in separate meter-in and separate meter-out control system. *Nonlinear Dynamics*, 1–20.

Chen, G., Wang, J., Wang, S., Zhao, J., Shen, W., & Li, J. (2016). Application of a new adaptive robust controller design method to electro-hydraulic servo system. *Acta Automatica Sinica*, *42*(3), 375–384.

Chiang, H.-K., & Tseng, C.-H. (2004). Design and implementation of a grey sliding mode controller for synchronous reluctance motor drive. *Control Engineering Practice*, *12*(2), 155–163.

Escareno, J.-A., Rakotondrabe, M., & Habineza, D. (2015). Backstepping-based robust-adaptive control of a nonlinear 2-DOF piezoactuator. *Control Engineering Practice*, *41*, 57–71.

Ginoya, D., Shendge, P., & Phadke, S. (2014). Sliding mode control for mismatched uncertain systems using an extended disturbance observer. *IEEE Transactions on Industrial Electronics*, *4*(61), 1983–1992.

Guan, C., & Pan, S. (2008). Nonlinear adaptive robust control of single-rod electro-hydraulic actuator with unknown nonlinear parameters. *IEEE Transactions on Control Systems Technology*, *16*(3), 434–445.

Hansen, A. H., Pedersen, H. C., Andersen, T. O., & Wachmann, L. (2011). Design of energy efficient smismo-els control strategies. In *proceedings of 2011 international conference on fluid power and mechatronics* (pp. 522–527).

Hao, R., Wang, J., Zhao, J., & Wang, S. (2016). Observer-based robust control of 6-dof parallel electrical manipulator with fast friction estimation. *IEEE Transactions on Automation Science and Engineering*, *13*(3), 1399–1408.

Jansson, A., & Palmberg, J.-O. (1990). *Separate controls of meter-in and meter-out orifices in mobile hyraulic systems, Technical report, SAE technical paper*.

Julong, D. (1989). Introduction to grey system theory. *The Journal of Grey System*, *1*(1), 1–24.

Krstic, M., Kanellakopoulos, I., & Kokotovic, P. V. (1995). *Nonlinear and adaptive control design*. Wiley.

Li, Z., Chen, J., Gan, M., Fang, H., & Zhang, G. (2010). Adaptive robust dynamic surface control of DC torque motors with true parameter estimates. In *Proceedings of the 2010 american control conference* (pp. 3524–3529). IEEE.

Liu, Y., Xu, B., Yang, H., & Zeng, D. (2009a). Calculation force control based on linearization feedback model of separate meter in and separate meter out control system. In *2009 IEEE international conference on automation and logistics* (pp. 1464–1468). IEEE.

Liu, Y., Xu, B., Yang, H., & Zeng, D. (2009b). Modeling of separate meter in and separate meter out control system. In *2009 IEEE/ASME international conference on advanced intelligent mechatronics* (pp. 227–232). IEEE.

Liu, Y., Xu, B., Yang, H., & Zeng, D. (2009c). Simulation of separate meter in and separate meter out valve arrangement used for synchronized control of two cylinders. In *2009 IEEE/ASME international conference on advanced intelligent mechatronics* (pp. 1665–1670). IEEE.

Liu, Y., Xu, B., Yang, H., & Zeng, D. (2009d). Simulation of separate meter in and separate meter out valve arrangement used for synchronized control of two cylinders. In *2009 IEEE/ASME international conference on advanced intelligent mechatronics* (pp. 1665–1670). IEEE.

Liu, S., & Yao, B. (2002). *Energy-saving control of single-rod hydraulic cylinders with programmable valves and improved working mode selection, Technical report, SAE technical paper*.

Liu, S., & Yao, B. (2006). Automated onboard modeling of cartridge valve flow mapping. *IEEE/ASME Transactions on Mechatronics*, *11*(4), 381–388.

Liu, S., & Yao, B. (2008). Coordinate control of energy saving programmable valves. *IEEE Transactions on Control Systems Technology*, *16*(1), 34–45.

Lu, Y.-S. (2009). Sliding-mode disturbance observer with switching-gain adaptation and its application to optical disk drives. *IEEE Transactions on Industrial Electronics*, *56*(9), 3743–3750.

Mohammadi, A., Tavakoli, M., Marquez, H., & Hashemzadeh, F. (2013). Nonlinear disturbance observer design for robotic manipulators. *Control Engineering Practice*, *21*(3), 253–267.

Na, J., Ren, X., Herrmann, G., & Qiao, Z. (2011). Adaptive neural dynamic surface control for servo systems with unknown dead-zone. *Control Engineering Practice*, *19*(11), 1328–1343.

Pazelli, T. F., Terra, M. H., & Siqueira, A. A. (2011). Experimental investigation on adaptive robust controller designs applied to a free-floating space manipulator. *Control Engineering Practice*, *19*(4), 395–408.

Pi, Y., & Wang, X. (2010). Observer-based cascade control of a 6-DOF parallel hydraulic manipulator in joint space coordinate. *Mechatronics*, *20*(6), 648–655.

Qiu, Y., Liang, X., & Dai, Z. (2015). Backstepping dynamic surface control for an anti-skid braking system. *Control Engineering Practice*, *42*, 140–152.

Song, H., Zhang, T., Zhang, G., & Lu, C. (2014). Robust dynamic surface control of nonlinear systems with prescribed performance. *Nonlinear Dynamics*, *76*(1), 599–608.

Sun, L., Li, D., & Lee, K. Y. (2015). Enhanced decentralized PI control for fluidized bed combustor via advanced disturbance observer. *Control Engineering Practice*, *42*, 128–139.

Swaroop, D., Hedrick, J. K., Yip, P. P., & Gerdes, J. C. (2000). Dynamic surface control for a class of nonlinear systems. *IEEE Transactions on Automatic Control*, *45*(10), 1893–1899.

Yang, J., Chen, W.-H., & Li, S. (2011). Non-linear disturbance observer-based robust control for systems with mismatched disturbances/uncertainties. *IET Control Theory & Applications*, *5*(18), 2053–2062.

Yang, J., Li, S., & Yu, X. (2013). Sliding-mode control for systems with mismatched uncertainties via a disturbance observer. *IEEE Transactions on Industrial Electronics*, *60*(1), 160–169.

Yao, B., Bu, F., & Chiu, G. T. (2001). Non-linear adaptive robust control of electro-hydraulic systems driven by double-rod actuators. *International Journal of Control*, *74*(8), 761–775.

Yao, B., Bu, F., Reedy, J., & Chiu, G.-C. (2000). Adaptive robust motion control of single-rod hydraulic actuators: theory and experiments. *IEEE/ASME Transactions on Mechatronics*, *5*(1), 79–91.

Yao, B., & DeBoer, C. (2002). Energy-saving adaptive robust motion control of single-rod hydraulic cylinders with programmable valves. In *Proceedings of the 2002 american control conference (IEEE Cat. No. CH37301) (Vol. 6)* (pp. 4819–4824). IEEE.

Yao, J., Jiao, Z., & Ma, D. (2014a). Adaptive robust control of DC motors with extended state observer. *IEEE Transactions on Industrial Electronics*, *61*(7), 3630–3637.

Yao, J., Jiao, Z., & Ma, D. (2014b). Extended-state-observer-based output feedback nonlinear robust control of hydraulic systems with backstepping. *IEEE Transactions on Industrial Electronics*, *61*(11), 6285–6293.

Yao, B., & Liu, S. (2002). Energy-saving control of hydraulic systems with novel programmable valves. In *Intelligent control and automation, 2002. Proceedings of the 4th world congress on (Vol. 4)* (pp. 3219–3223). IEEE.

Yao, B., & Palmer, A. (2002). Indirect adaptive robust control of SISO nonlinear systems in semi-strict feedback forms. *IFAC Proceedings Volumes*, *35*(1), 397–402.

Yao, B., & Tomizuka, M. (1997). Adaptive robust control of siso nonlinear systems in a semi-strict feedback form. *Automatica*, *33*(5), 893–900.

Zeng, B., Liu, S., & Xie, N. (2010). Prediction model of interval grey number based on DGM (1, 1). *Journal of Systems Engineering and Electronics*, *21*(4), 598–603.